Tool Names Versus Clarification and Refusal Text

Status: Owner-run component measurement completed. Date: 2026-10-15.

Question

Whole-answer loss is higher for CLARIFY and REFUSE than ACT. Locate that difference within the answer: choosing the tool name, composing boundary text, or other tokens. Do not change the checkpoint, prompt, dataset, labels, caps, or benchmark rules.

Command

```
.venv\Scripts\python scripts\report_sft_loss.py `
  --split train `
  --max-rows 1000 `
  --max-sequence-length 1024 `
  --by-component `
  --output tmp\name-first-training-loss-components.json
```

The default remains the same frozen two-epoch QKVO name-first checkpoint. The output is a new artifact; the earlier whole-answer report remains untouched. Without `--by-component`, report behavior and schema remain unchanged.

Components

Component	Scored content
`first_tool_name`	The first call's `name` string contents.
`later_tool_name`	Other call-name string contents in multi-tool targets.
`question_text`	The `arguments.question` string contents of `request_clarification`.
`reason_text`	The `arguments.reason` string contents of `refuse_request`.
`other`	JSON syntax, keys, productive-tool arguments, and visible chat endings.
`mixed_boundary`	Tokens overlapping both a named string region and its surrounding syntax.

Use the standard JSON decoder to locate field values without reserializing any target. Fast-tokenizer character offsets map those regions onto the exact training tokens. Quotes are not part of the string-content regions. Tokens crossing a region boundary are not silently attributed to pure names or wording. Escaped text remains in its original serialized representation.

Verify offset-tokenized IDs match training IDs and that the prompt token sequence is a prefix of the full input. Prompt labels and position zero remain excluded. Preserve skipped and partially truncated rows exactly as the original diagnostic.

One forward pass supplies the model loss and logits. Compute unreduced cross-entropy only on supervised positions, using each token's preceding logit. Select at most 64 supervised positions at a time to limit temporary FP32 memory. Reject non-finite losses and discrepancies between component totals and the model's reported loss. Every scored token belongs to exactly one component.

The console adds a class/component table with contributing rows, tokens and loss. The JSON includes per-row components, class summaries and all-class component summaries, alongside the existing identity hashes and whole-answer losses.

Interpretation

Large first-name loss supports investigating tool-choice learning. Low name loss alongside higher question/reason loss supports investigating supplied-answer text. Neither finding by itself proves a cause or chooses the next training intervention.

These are teacher-forced losses, not free-generation routing probabilities or semantic adequacy scores. Later tool-name pieces see earlier gold name pieces; question and reason tokens see the correct gold tool name. In argument-first formats, even the first name sees preceding gold arguments, so format must remain fixed when interpreting comparisons. Inspect mixed-boundary coverage before interpreting pure-string statistics.

Owner-Run Result

Source: `tmp/name-first-training-loss-components.json` and the owner's terminal output. Whole-answer losses and retained row counts matched the earlier class-loss report.

Class	Component	Rows	Tokens	Loss/token
ACT	First tool name	774	2,575	0.016015
ACT	Later tool names	338	1,410	0.001974
ACT	Other	774	26,879	0.043888
CLARIFY	First tool name	100	400	0.276975
CLARIFY	Question text	100	1,045	0.817933
CLARIFY	Other	100	1,300	0.000393
CLARIFY	Mixed boundary	100	100	0.193296
REFUSE	First tool name	100	300	0.164804
REFUSE	Reason text	100	1,205	0.651727

Class	Component	Rows	Tokens	Loss/token
REFUSE	Other	100	1,300	0.000294
REFUSE	Mixed boundary	100	100	0.351888

Boundary JSON syntax is fitted closely; the largest boundary-answer loss is on supplied question/reason text. Different acceptable phrasing can also incur text loss, so this is not proof of inadequate questions or reasons. First meta-tool names have higher loss than productive first names, but name-loss averages still condition on earlier gold name pieces and are not a routing confusion matrix.

The owner chose a first-tool-name weighting ablation before considering a router/generator architecture change. This is a routing-focused hypothesis, not a demonstrated remedy for the higher text loss.

No checkpoint was run by the coding agent. Verification uses mocked inputs, tiny CPU tensor arithmetic when PyTorch is installed, and model-free preflight.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.